Session 12: AlphaFold and Generative Models of Protein Structure

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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Residues that touch tend to mutate together over evolution. What is this called?

A Conservation

B Homology

C Coevolution

D Convergent evolution

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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A region of an AlphaFold model has pLDDT 30. What does that mean?

A 30% of its atoms are wrong

B It is a β-strand

C It is predicted very accurately

D AlphaFold is unsure of it; often it is disordered

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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How does a diffusion model generate a new example?

A It copies the nearest training example

B It starts from noise and removes it step by step

C It folds the chain with a force field

D It searches the PDB for a match

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Contacts	AlphaFold2	Complexes, design
coevolution, contact maps	Evoformer, pLDDT and PAE	interfaces, diffusion

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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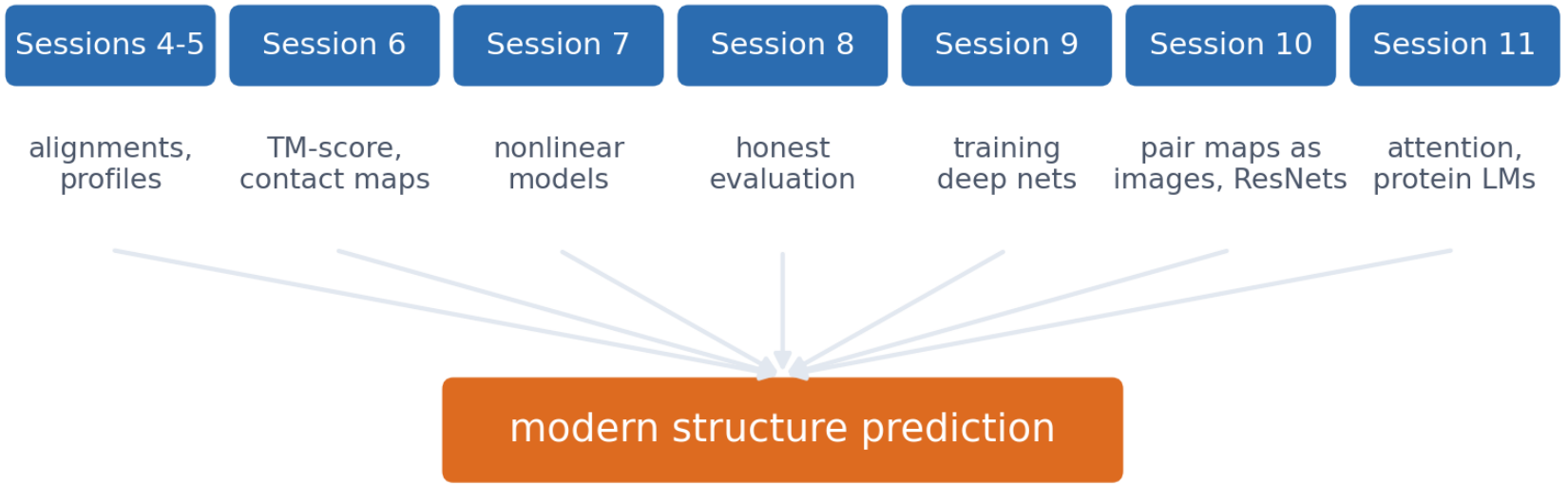
What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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Every session contributes a piece



The question of the day: what information is missing if we predict a residue's structure from its local sequence alone? Contacts with residues far away in the sequence.

See course.bioinfo.se/session12.html, introduction.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: where long-range structural information comes from. Block 2: how AlphaFold2 combines the course's ideas, and what its confidence means. Block 3: complexes, and generative models.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

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BLOCK 1 · ABOUT 30 MIN

Where does long-range structural information come from?

From a line of letters to a 3D object

The sequence: 146 amino acids

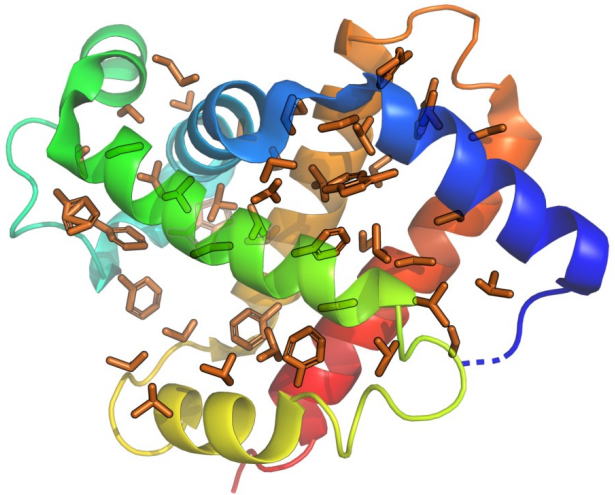
V H L T P E E K S A V T A L W G K V N V D
E V G G E A L G R L L V V Y P W T Q R F F
E S F G D L S T P D A V M G N P K V K A H
G K K V L G A F S D G L A H L D N L K G T
F A T L S E L H C D K L H V D P E N F R L
L G N V L V C V L A H H F G K E F T P P V
Q A A Y Q K V V A G V A N A L A H K Y H

orange: hydrophobic (A, V, L, I, M, F, W)

easy to measure: about 150 million in UniProt

prediction

The structure: the same chain in 3D



blue = N-terminus ... red = C-terminus; orange sticks: hydrophobic side chains, mostly buried in the core. Hard to measure: about 260,000 structures in the PDB

A chain of amino acids folds because hydrophobic side chains (orange) are buried away from water, the backbone forms hydrogen bonds (helices, strands) and the core packs tightly. Sequences are easy to measure (about 150 million in UniProt), structures hard (about 260,000 in the PDB): fewer than 1 in 500.

PDB 4HHB chain B, rendered with PyMOL; tools/builders/make_session12_intro.py. See course.bioinfo.se/session12.html, 'The starting point'.

Two classic observations

Anfinsen

the sequence decides

- Ribonuclease unfolded with urea and a reducing agent
- Refolds on its own, with the right 4 of 105 possible disulphide pairings
- So prediction is possible in principle
- Not for all: chaperones, partners, switches, disorder

Levinthal

$3^{100} \approx 5 \times 10^{47}$ conformations

- At 10^{13} per second: 5×10^{34} s to try them all
- Age of the universe: 4×10^{17} s
- Real proteins fold in µs to s: a guided funnel, not a random search
- **Lesson: every method needs a smarter search**

Anfinsen (1973), Science 181:223; Levinthal (1969). See course.bioinfo.se/session12.html.

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Before deep learning: four ways to predict a structure

Homology modelling

copy a relative

- Align to a template (Sessions 4-5)
- Works above ~30% identity
- No template: nothing to copy

Threading

find a fold that fits

- Thread the sequence onto every known fold
- Score burial and contacts
- Finds templates sequence search misses

Fragment assembly

Rosetta

- 3- and 9-residue pieces of known structures
- Monte Carlo + knowledge-based energy
- New folds, but small proteins only

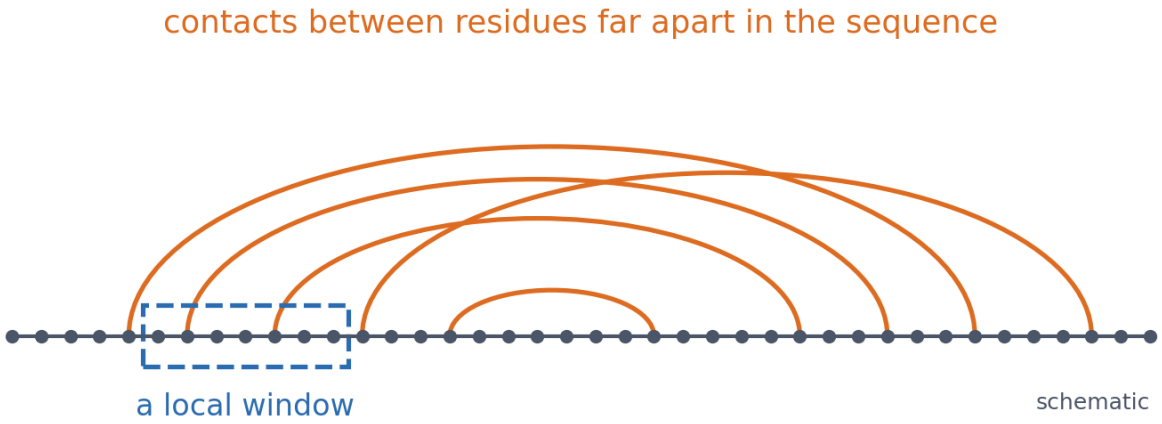
Molecular dynamics

simulate the physics

- Newton's laws, 2 fs steps
- Anton folded 12 small proteins (2011)
- Far too slow for ordinary proteins

Šali & Blundell 1993; Jones, Taylor & Thornton 1992; Simons et al. 1997; Lindorff-Larsen et al. 2011. See course.bioinfo.se/session12.html, 'How structures were predicted before deep learning'.

Local sequence is not enough



- **Same ten residues: helix or strand**
- (Session 7's chameleons)
- **What is missing: contacts far away**

See course.bioinfo.se/session12.html, 'The starting point'.

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Three generations of structure prediction



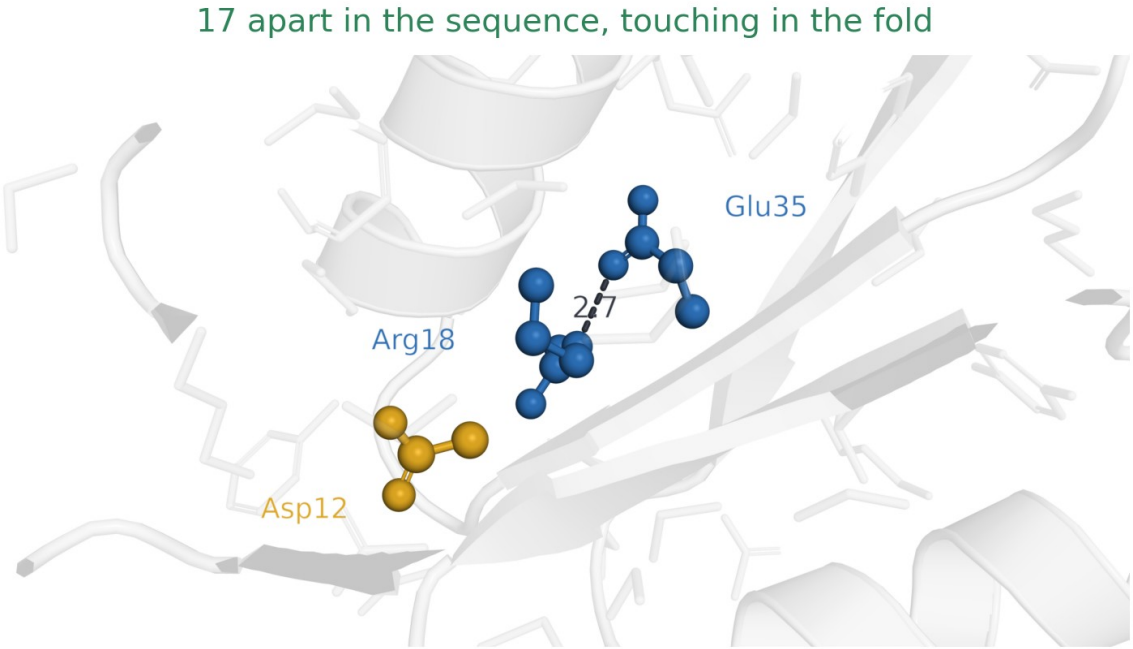
CASP blind tests every two years since 1994

Physics and templates gave steady progress for proteins with a template and little for the rest. What all early approaches lacked was long-range information.

Dill & MacCallum (2012), Science 338:1042; Moult et al. (1995).

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Why residues in contact change together

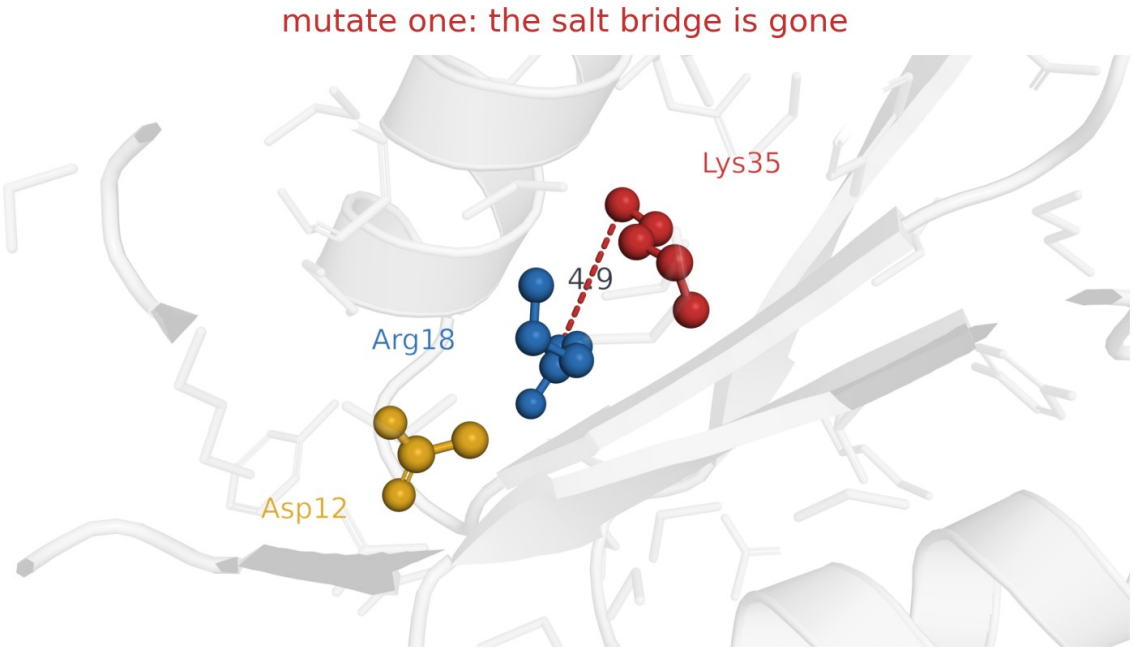


Arg18 and Glu35 sit 17 apart in the sequence and form a buried salt bridge: one positive, one negative, 2.7 Å apart. Blue is the residue as CheY actually has it; gold is Asp12, which comes back at the end. Every other side chain is drawn faintly, to show how tightly packed the neighbourhood is.

CheY (PDB 3CHY), the course's running example for contact prediction.

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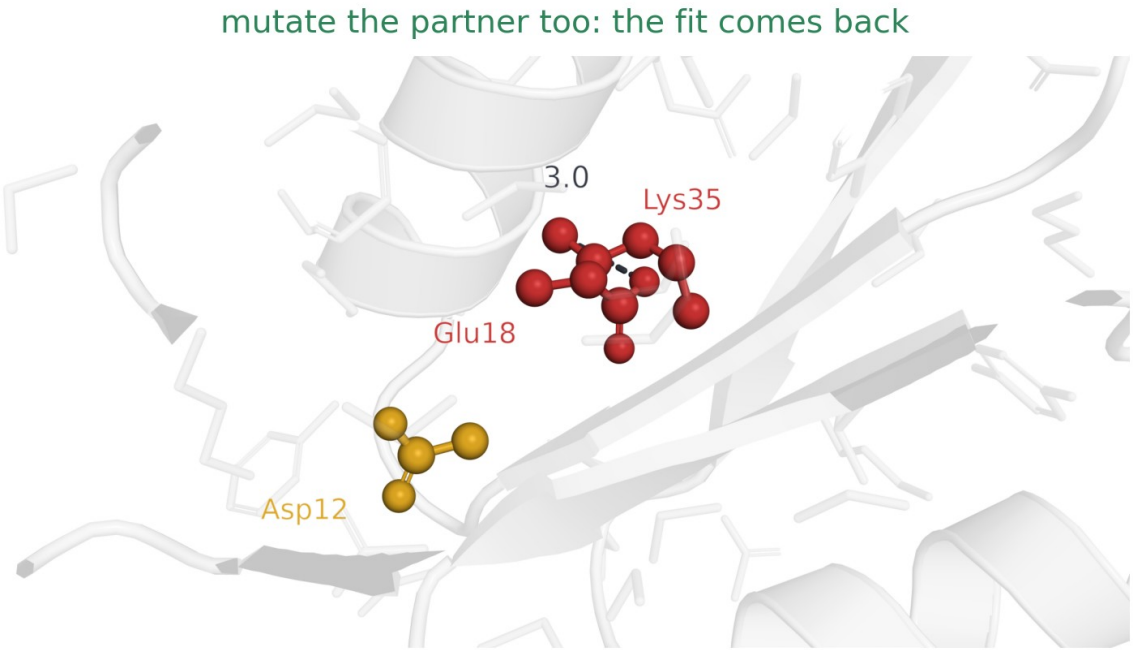
Why residues in contact change together



Replace Glu35 with a lysine — red, because it is the one that has been swapped — and the partner is still positive: 4.9 Å, and two like charges where there was a salt bridge. The structure is destabilised, and selection removes the variant.

CheY (PDB 3CHY), the course's running example for contact prediction.

Why residues in contact change together

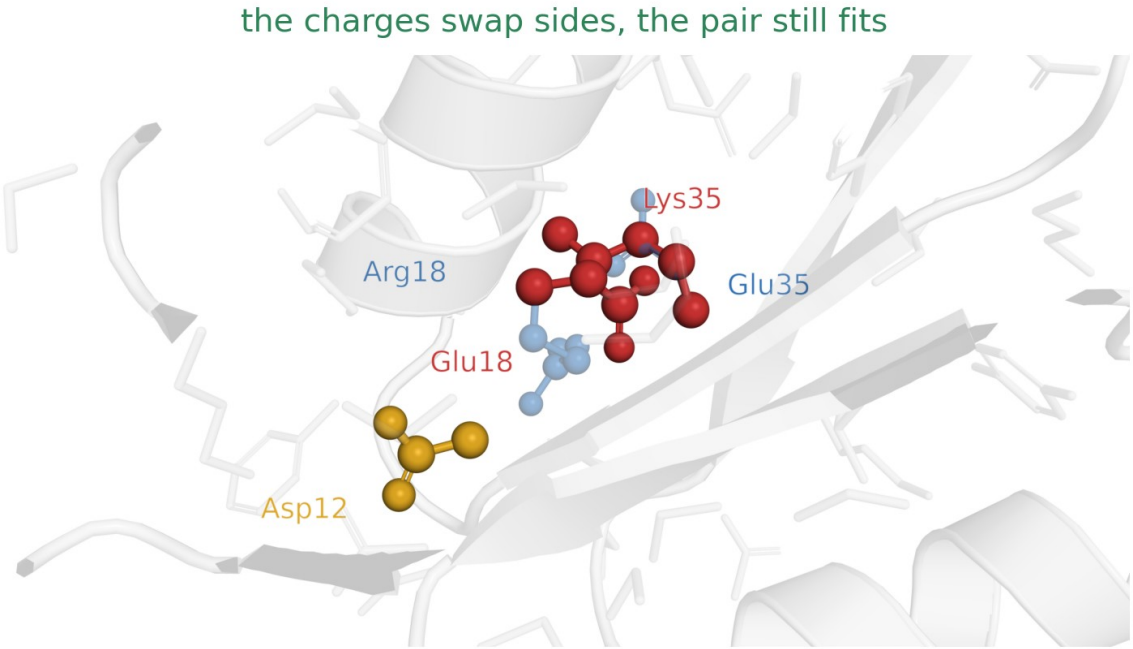


Now change the partner as well, Arg18 to a glutamate. Both positions are red, both have been swapped, the charges have changed places, the salt bridge is back at 3.0 Å, and the double mutant is as fit as the original.

CheY (PDB 3CHY), the course's running example for contact prediction.

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Why residues in contact change together

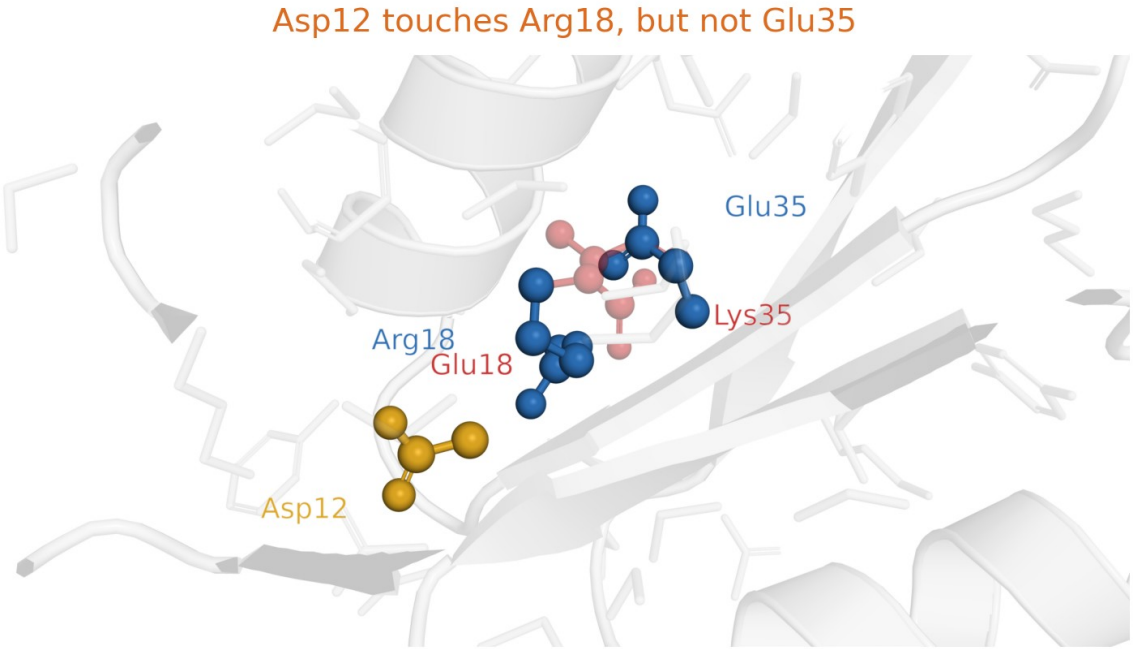


The wild type laid back over the double mutant: pale blue is Arg18 and Glu35 as CheY has them, solid red the glutamate and lysine that replaced them. The charges have changed places, the pair still fits — and a third residue has been sitting there, in gold, since the first slide.

CheY (PDB 3CHY), the course's running example for contact prediction.

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Why residues in contact change together



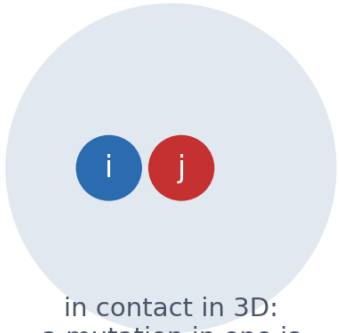
Back to the real CheY, with Asp12 in the gold it has had all along: it touches Arg18, and Arg18 touches Glu35, but Asp12 and Glu35 never touch — 8.2 Å between Cβ atoms, against 5.1 Å and 4.9 Å for the two real contacts. Both are nonetheless constrained by Arg18, so in an alignment they appear coupled. The double mutant is still there as the faint red copy behind: whichever residue sits at 18, it is the one Asp12 is pressed against.

CheY (PDB 3CHY), the course's running example for contact prediction.

Residues in contact change together

	i							j	
M	K	V	L	E	A	G	T	E	L
M	R	V	L	D	A	G	S	D	L
M	K	I	L	E	A	G	T	D	V
M	E	V	L	K	A	G	T	K	L
M	D	V	L	R	A	G	S	R	L
M	E	I	L	K	A	G	T	R	V

toy alignment: when i is +, j is −, and vice versa



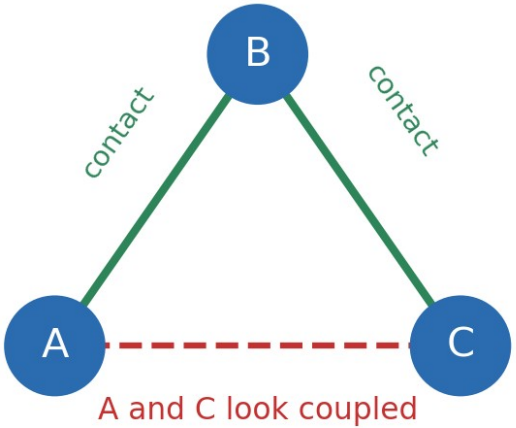
in contact in 3D:
a mutation in one is
compensated in the other

- **A mutation in one is compensated in the other**
- Visible across a large family
- 1994: Göbel, Sander, Schneider & Valencia; Shindyalov, Kolchanov & Sander
- Real signal, but too weak to fold a protein
- A Potts model; fitted by mean field, or plmDCA/GREMLIN

Göbel et al. (1994), Proteins 18:309; Shindyalov et al. (1994), Protein Eng. 7:349. See course.bioinfo.se/session12.html, 'Coevolution'.

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Correlation is fooled by indirect effects



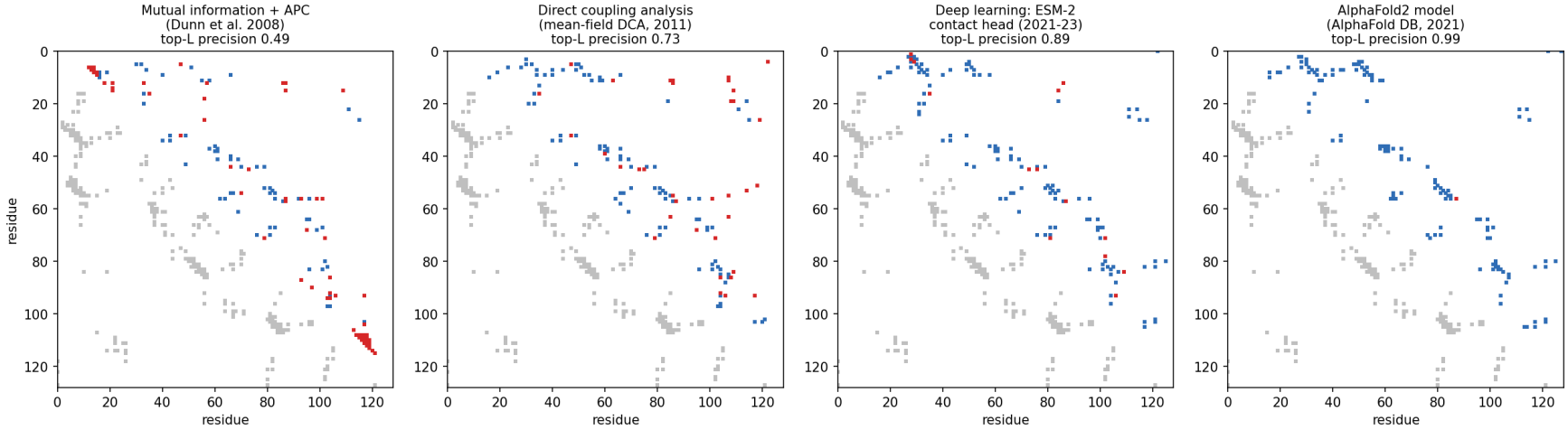
correlation: A-C false positive

DCA: one global model
keeps only direct couplings

- **A-B and B-C contacts**
- make A and C look coupled
- **DCA: one global model**
- Shown by Lapedes et al. in 1999, but barely noticed

Lapedes et al. (1999), IMS Lect. Notes 33:236; Lapedes, Giraud & Jarzynski (2002), arXiv:1207.2484; Weigt et al. (2009); Morcos et al. (2011), PNAS 108:E1293.

From mutual information to deep learning: CheY

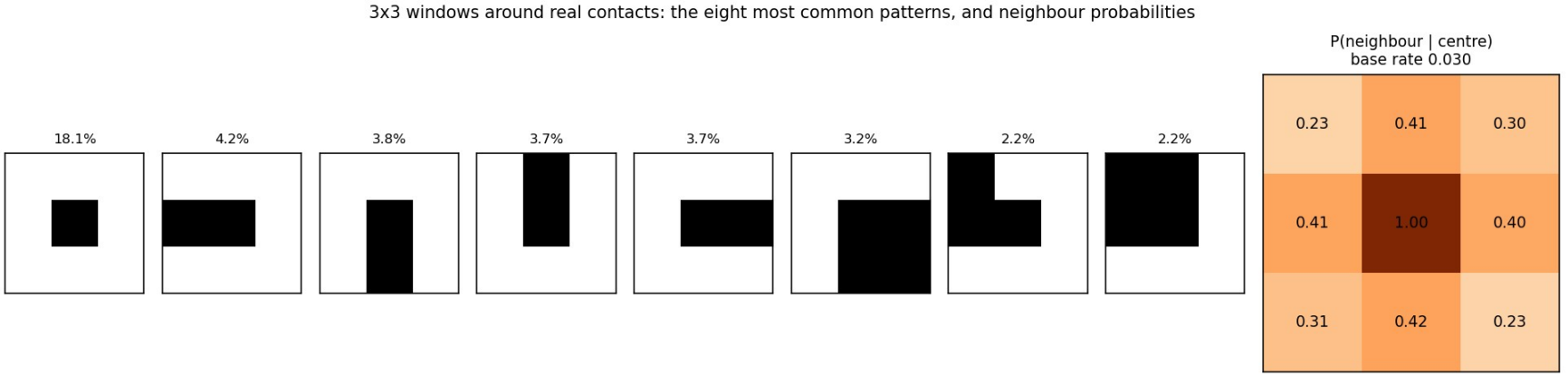


Same 11,084-sequence alignment; the top 128 predicted contacts (blue true, red false) over the real map. Precision: mutual information 0.49, DCA 0.73, deep learning (ESM-2 contact head) 0.89, AlphaFold2 model 0.99.

Notebook, Part 1; Dunn 2008; Morcos 2011; Rao 2021; Jumper 2021.

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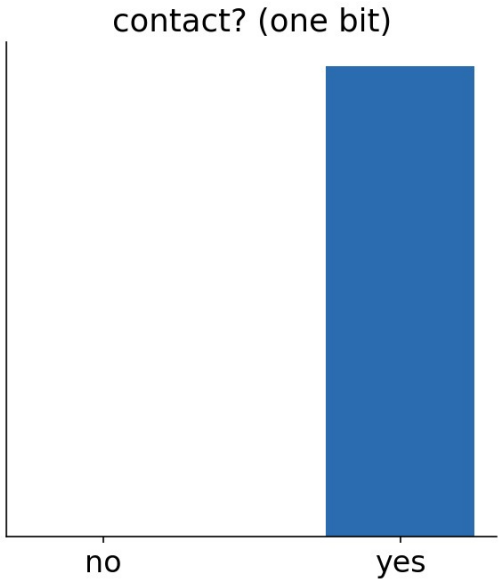
Contacts are not independent pixels

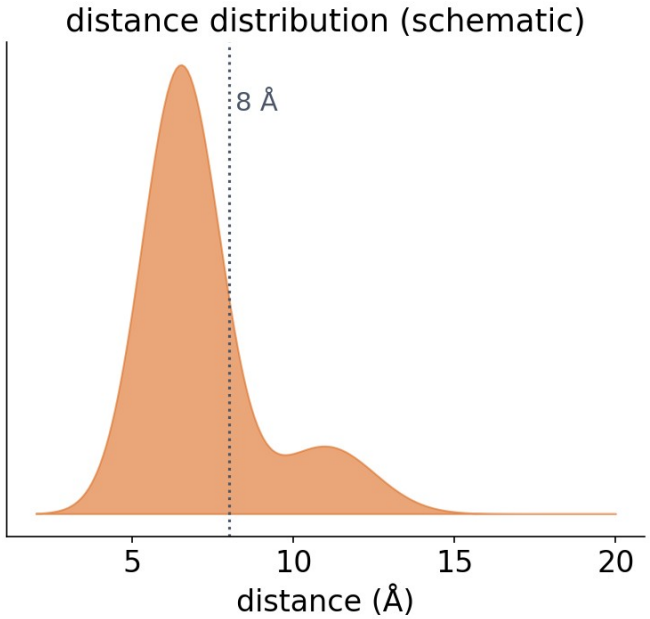


3 × 3 windows around real contacts in eight structures: only 146 of the 256 possible patterns occur. If the centre pair is in contact, a neighbouring pair is in contact 23-42% of the time, against a base rate of 3.0%.

Notebook, Part 1.

From contacts to distances



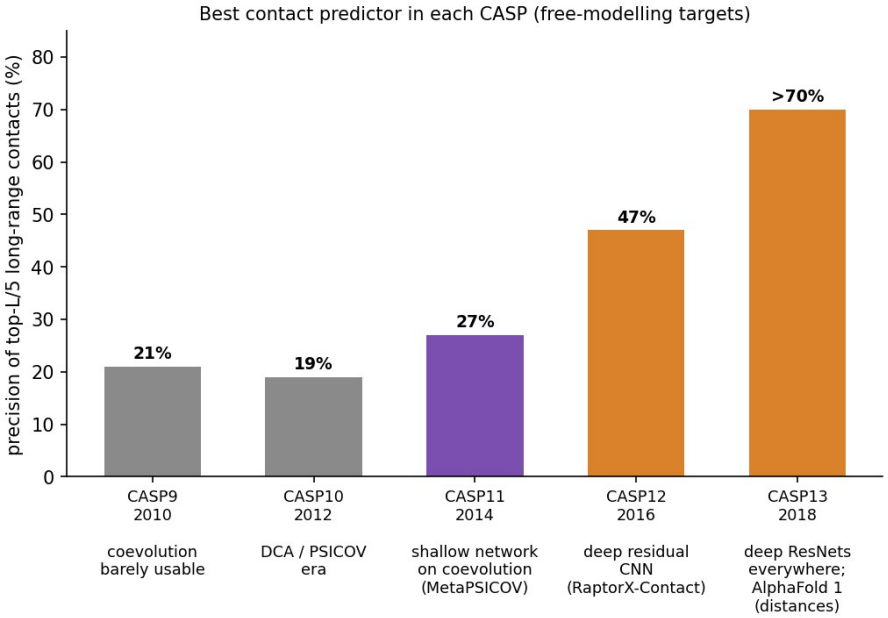


- **Predict a distance distribution**
- per residue pair
- Fold by minimising against it
- AlphaFold 1, trRosetta (2018-20)

Senior et al. (2020), Nature 577:706; Yang et al. (2020), PNAS 117:1496.

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CASP: a sudden change



CASP14 (2020): top-5 average 64%, no further gain; AlphaFold 2 made contact prediction a by-product of full 3D prediction.

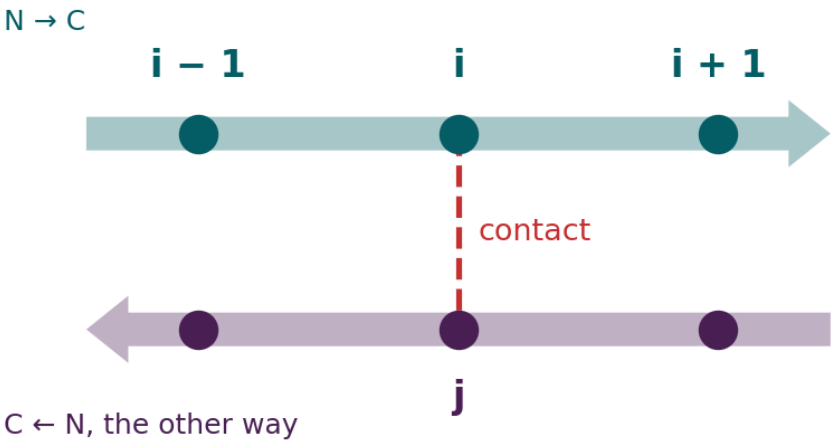
- **2010-2014: 20-27%**
- 2016: 47%
- **2018: more than 70%**
- 2020: contacts a by-product of AlphaFold2

Official CASP assessment papers (CASP10-CASP14).

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Which neighbouring pair is in contact too?

Two antiparallel β-strands; residue i is in contact with residue j. The strands run in opposite directions.



A (i + 1, j + 1): along the diagonal

B (i + 1, j − 1): across the diagonal

C (i + 1, j) and (i, j + 1) most

D All neighbours equally

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Which neighbouring pair is in contact too?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Which neighbouring pair is in contact too?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session12-discussion-1.ipynb

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Which neighbouring pair is in contact too?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Which neighbouring pair is in contact too?

Results, and who explains

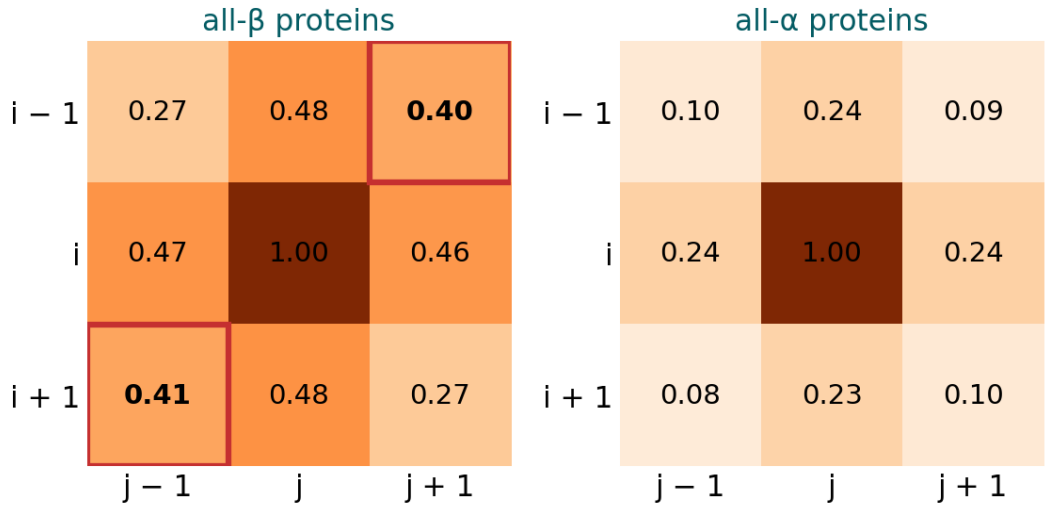
Both rounds side by side.

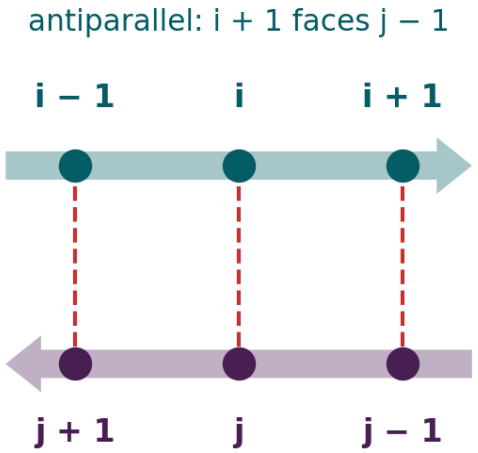
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: (i + 1, j − 1), across the diagonal





Three all-β and three all-α structures, every contact with j − i ≥ 12. In the β-proteins, (i + 1, j − 1) is a contact 40% of the time, (i + 1, j + 1) 27%: antiparallel strands run in opposite directions, so i + 1 faces j − 1. Helices give a weaker neighbourhood, since only every third or fourth residue faces the partner. A predictor that looks at a patch of the map can use this: PconsC2, the CNNs, AlphaFold2's pair updates.

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BLOCK 1 DONE

Break

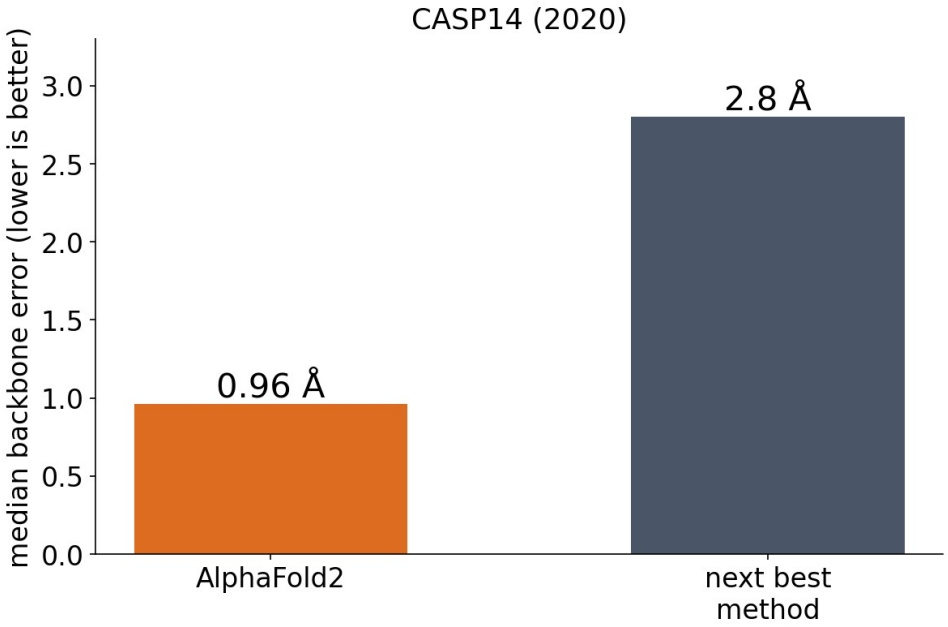
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BLOCK 2 · ABOUT 30 MIN

How AlphaFold2 combines the course's ideas

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Atomic accuracy, even without a similar structure

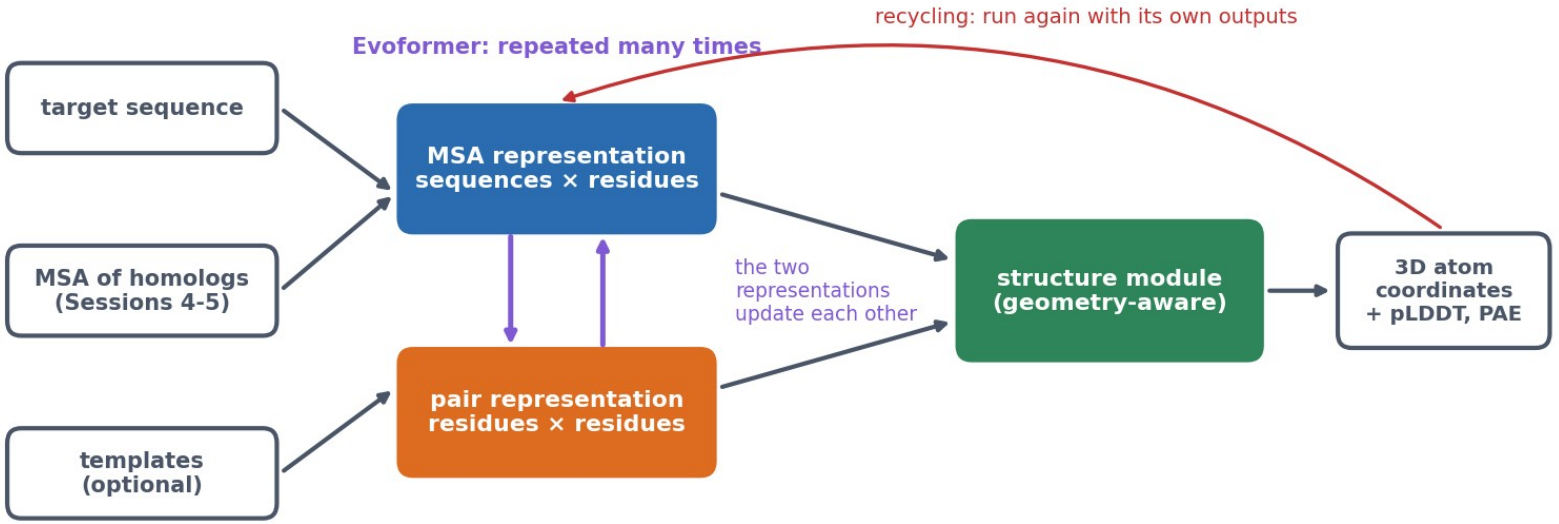


- **Trained end to end**
- Sequence, alignment, templates → atoms
- Unlike ESMFold: a single sequence

Jumper et al. (2021), Nature 596:583.

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AlphaFold2: the data flow

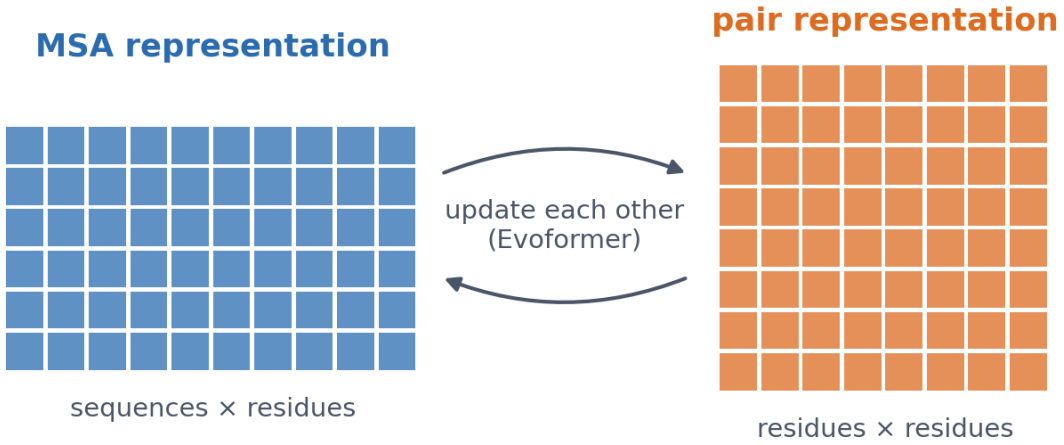


learned coevolution (MSA → pair), triangle updates within the pair map, rotation-invariant geometry in the structure module

Inputs are embedded into an MSA representation and a pair representation, which the Evoformer updates many times. A geometry-aware structure module builds 3D coordinates with confidence scores; recycling runs the network again on its own outputs.

Jumper et al. (2021).

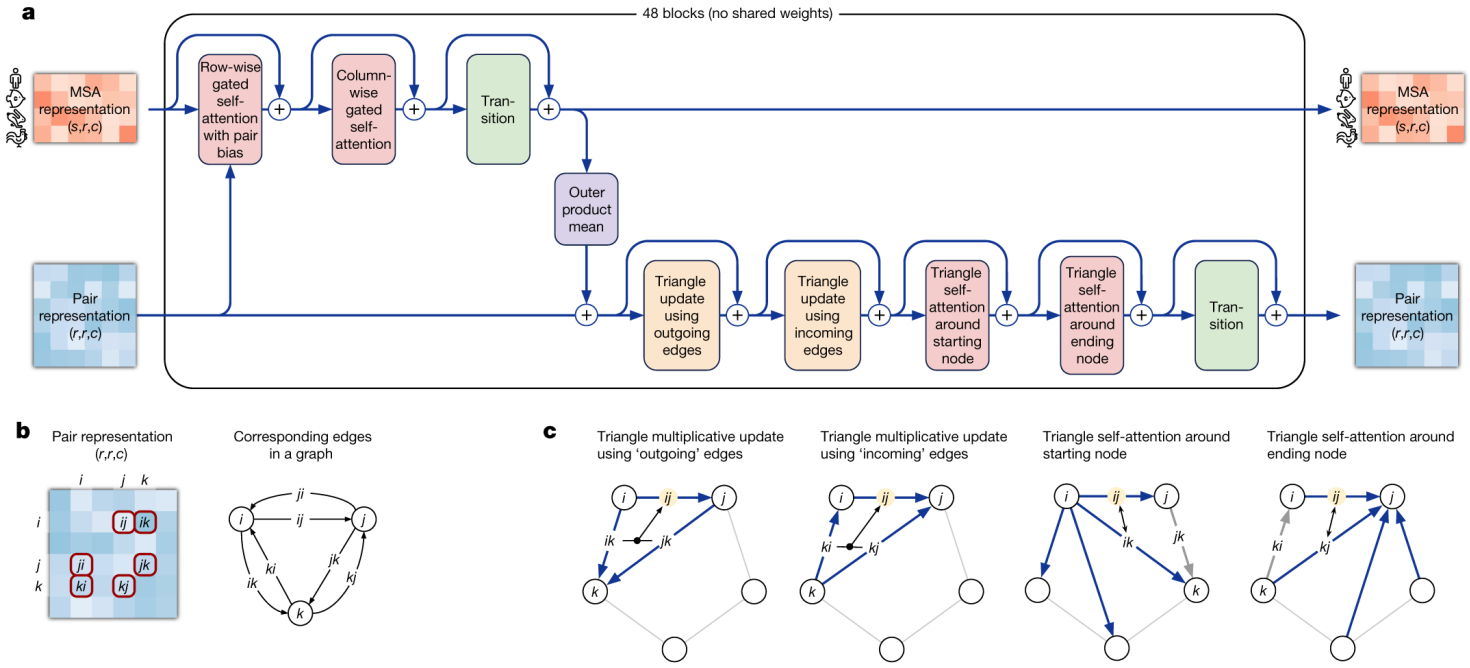
Two representations



- **MSA: a vector per residue per sequence**
- **Pair: a vector per residue pair**
- A richer contact map

See course.bioinfo.se/session12.html, 'The data flow'.

One Evoformer block, repeated 48 times



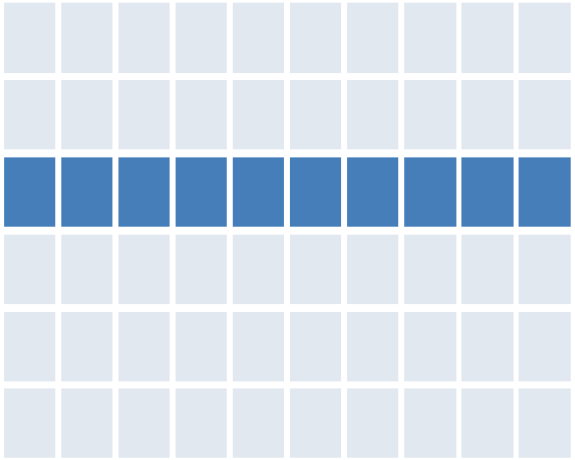
(a) MSA row (top): row attention biased by the pair, column attention, transition. Outer product mean: MSA → pair. Pair row (bottom): four triangle operations, transition. Every ⊕ is a residual connection. (b, c) The pair as graph edges; each triangle update combines edges via a third residue k.

Jumper et al. (2021), Nature 596:583, Fig. 3a-c (cropped). CC BY 4.0.

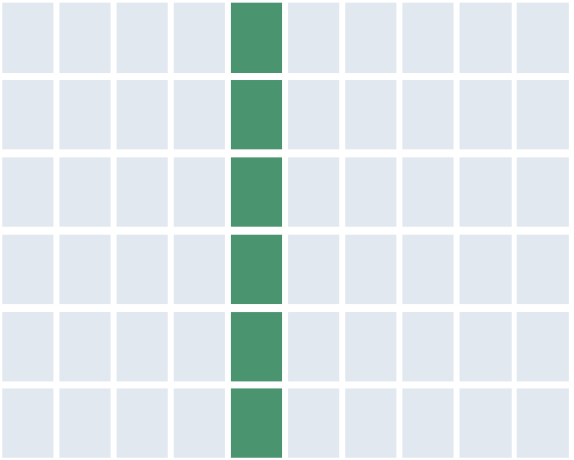
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MSA attention: rows and columns

within a sequence (rows):
which residues give context?



within a column:
which homologs are informative?

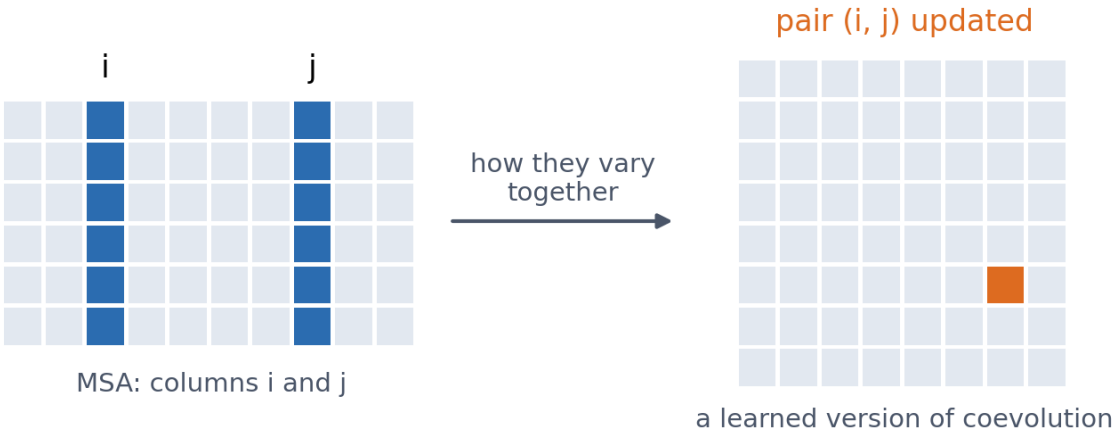


Within each aligned sequence, residues attend to each other (Session 11's attention, biased by the pair representation). Within each column, sequences attend to each other.

Jumper et al. (2021).

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MSA → pair: learned coevolution

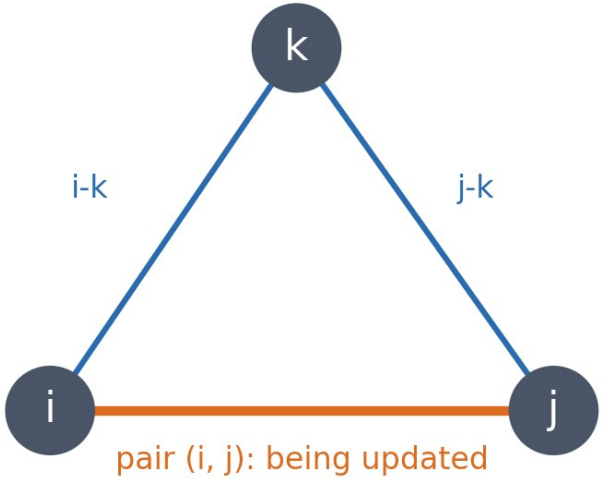


- **How two columns vary together**
- updates pair (i, j)
- Block 1, as a learned operation

Jumper et al. (2021): the outer product mean.

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Triangle updates



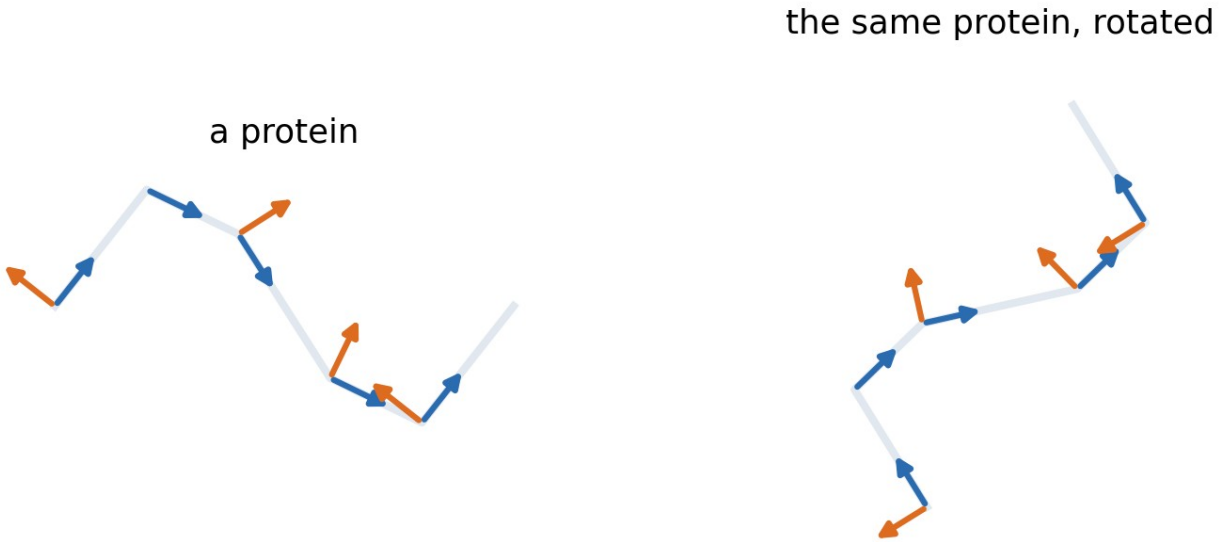
for every
third residue k

- **(i, j) informed via every k**
- Pairs are not independent
- Inspired by the triangle inequality
- **Learned, not enforced**

Jumper et al. (2021).

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Residue frames, and invariance in 3D

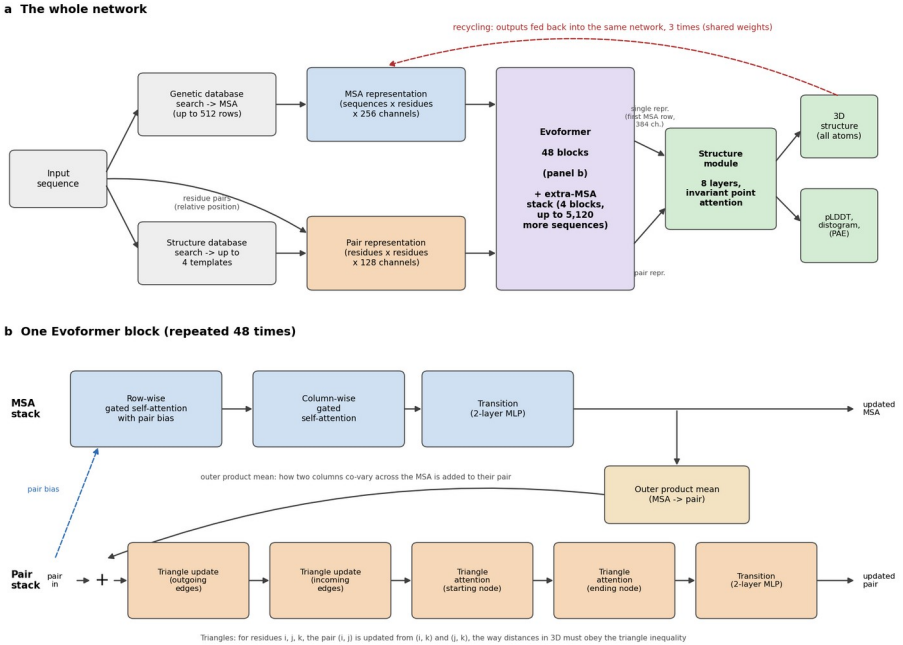


each residue is a frame (position + orientation); the answer does not depend on the rotation

Each residue is a rigid frame (backbone position and orientation). Invariant point attention does not depend on how the whole protein is rotated or translated: Session 10's invariance, in three dimensions. Recycling then refines the prediction a few times.

Jumper et al. (2021).

AlphaFold2 in detail



MSA representation (256 channels) and pair representation (128 channels) through 48 Evoformer blocks; an 8-layer structure module; 3 recycles. Panel b: the operations of one Evoformer block.

Drawn after Jumper et al. (2021) and its Supplementary Information.

SciLifeLab

pLDDT: how trustworthy is the local structure?

pLDDT (0-100, per residue): how trustworthy is the local structure here?

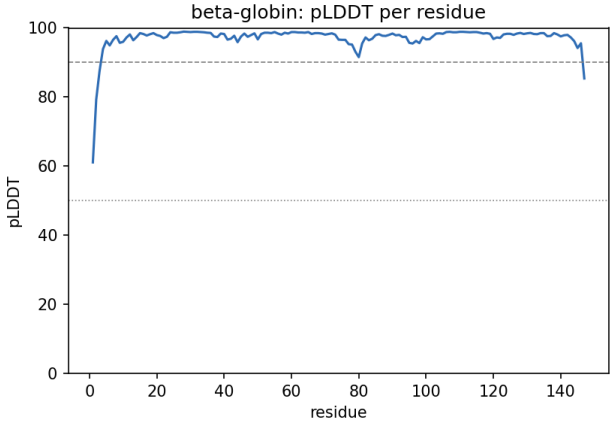


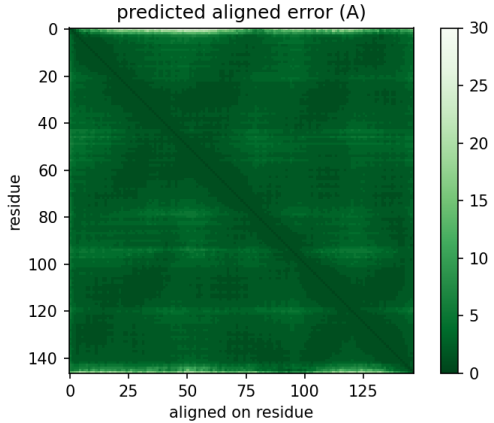
a confidence score is the model's own estimate, learned from its training data

A predicted structure is only useful if we know how far to trust it: the confidence scores are as important as the structure itself.

See course.bioinfo.se/session12.html, 'Confidence: pLDDT and PAE'.

PAE: aligned on A, how sure is B?

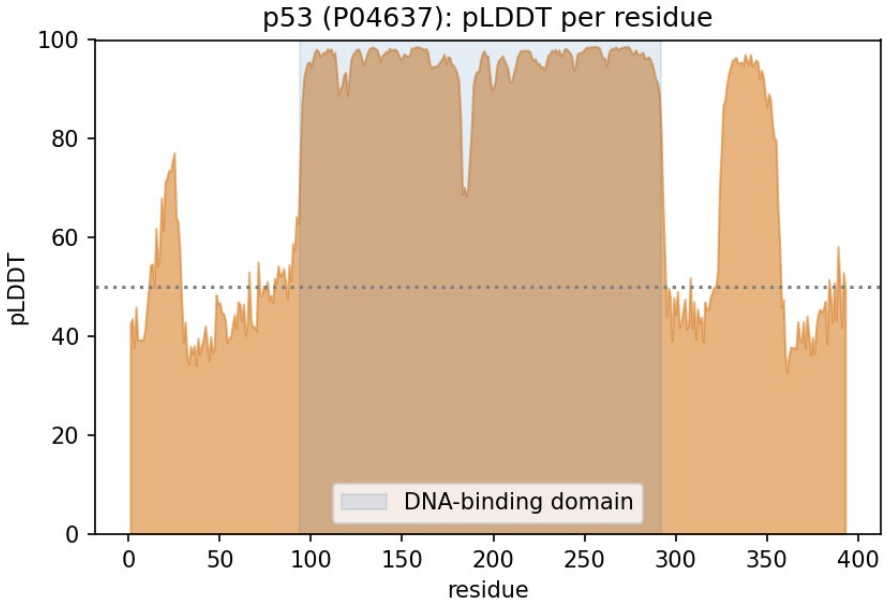


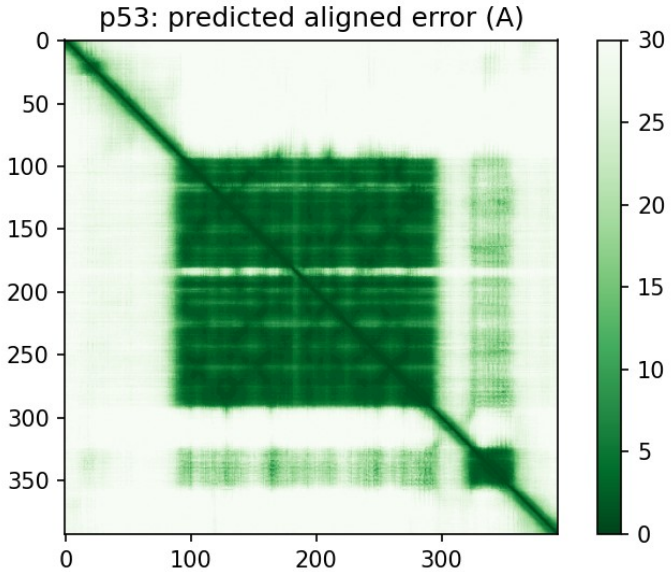


- **Per pair of residues**
- Domains right, orientation unknown
- **β-globin: pLDDT 97.2, TM 0.988**

Notebook, Part 2: AlphaFold DB model vs crystal structure 4HHB.

p53: one confident domain

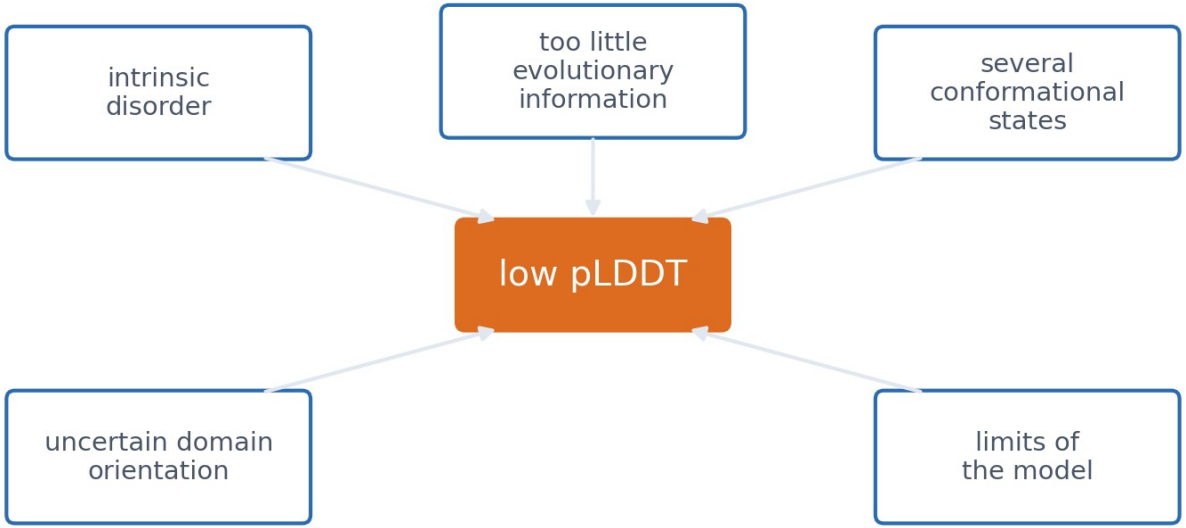




30% of the residues have pLDDT below 50. DNA-binding domain: mean 95.3; N-terminal transactivation region 47.6; C-terminus 45.8. Here low confidence matches biology: these regions are known to be intrinsically disordered.

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Low pLDDT is not the same as disorder



- **Confidence ≠ correctness**
- The model's own estimate
- Famous proteins: in the training data

See course.bioinfo.se/session12.html, 'Confidence: pLDDT and PAE'.

SciLifeLab

Structures for every protein

- **AlphaFold DB: essentially all of UniProt**
- ColabFold: MMseqs2 + AlphaFold in Colab
- ESMFold: one sequence, no alignment

0.941

TM-score, ESMFold
Ephrin-A5 (1SHW)

2024

Nobel Prize: Hassabis,
Jumper; Baker

Tunyasuvunakool et al. 2021; Varadi et al. 2024; Mirdita et al. 2022; Lin et al. 2023.

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What AlphaFold cannot tell you

Cannot tell

from one prediction

- whether a sequence folds at all
- how a mutation changes the structure

Often unreliable

for

- alternative conformational states
- domain arrangement in large flexible proteins
- membranes, protein-DNA complexes

From the lecture; see course.bioinfo.se/session12.html, 'Known limits'.

SciLifeLab

What is the mean pLDDT of the p53-binding domain?

MDM2, which binds p53. Full-length human MDM2, AlphaFold DB model: 491 residues, mean pLDDT 63.




```
plddt = np.array([r["CA"].get_bfactor()
                  for r in chain])
print(plddt.mean())         # 62.6
domain = plddt[24:109]  # residues 25-109
print(domain.mean())        # ?
```

A About 63: like the whole chain

B About 95: the rest pulls the mean down

C About 40: no part can be trusted

D Cannot say without a crystal structure

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

SciLifeLab

What is the mean pLDDT of the p53-binding domain?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What is the mean pLDDT of the p53-binding domain?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session12-discussion-2.ipynb

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What is the mean pLDDT of the p53-binding domain?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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What is the mean pLDDT of the p53-binding domain?

Results, and who explains

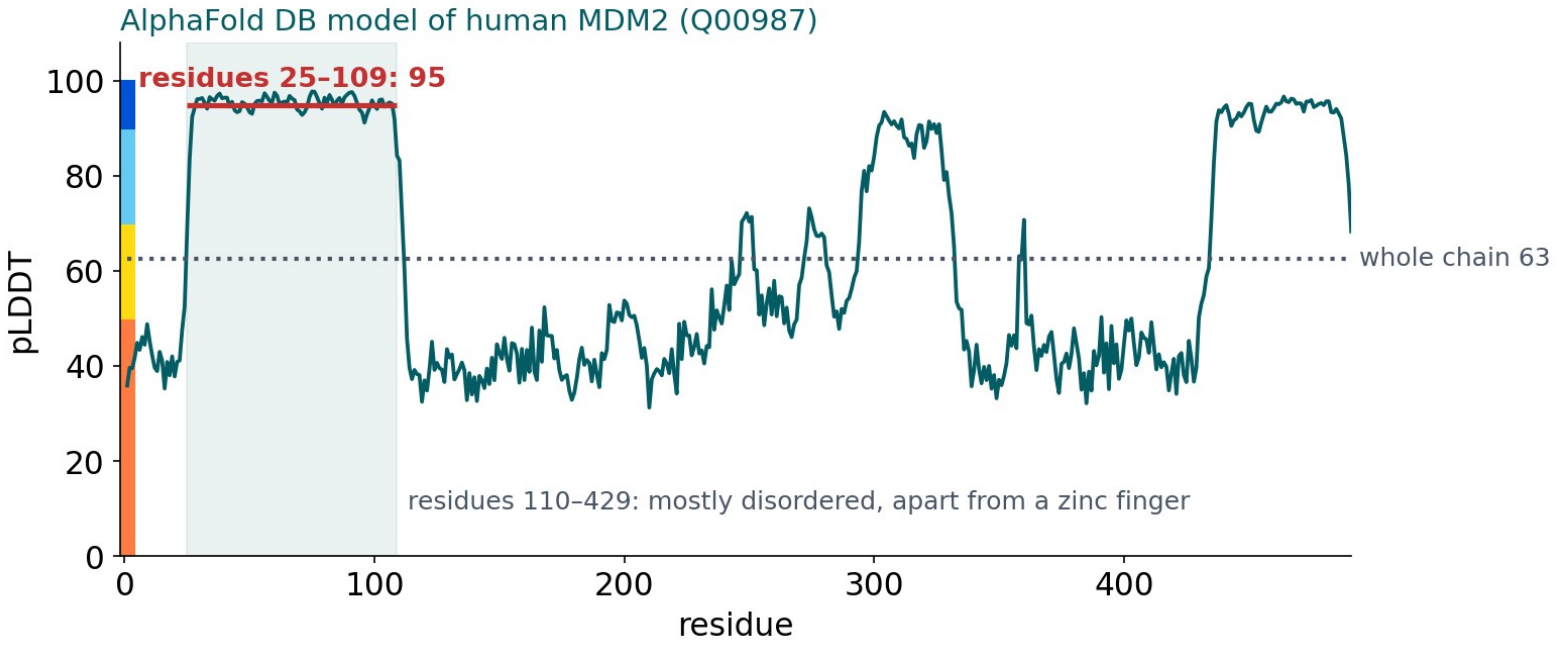
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: about 95; the mean hides a confident domain



The p53-binding domain of MDM2 has a mean pLDDT of 95; residues 110–429 average 50. A mean of 63 is not a mediocre model everywhere: it is confident domains joined by long stretches AlphaFold does not place, in MDM2 mostly disordered, as in p53. Look at pLDDT along the chain, and at the PAE between domains, never only at the mean.

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BLOCK 2 DONE

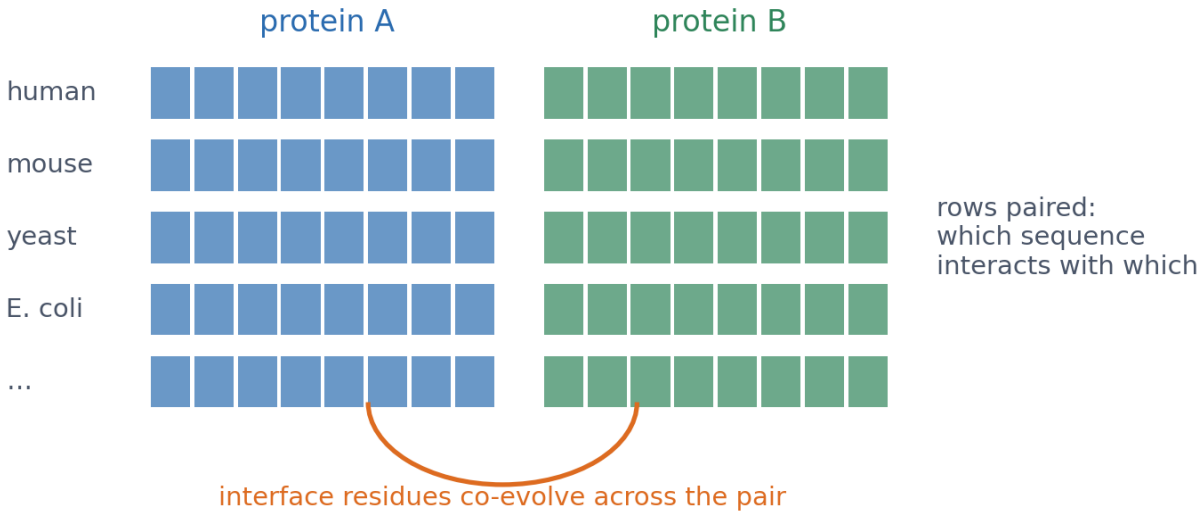
Break

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BLOCK 3 · ABOUT 30 MIN

From single chains to interactions, and generative models

Complexes: pair the alignments



- **Interfaces co-evolve too**
- FoldDock: AlphaFold2 on joined chains
- AlphaFold-Multimer: retrained

Bryant, Pozzati & Elofsson (2022), Nat Commun 13:1265; Evans et al. (2021).

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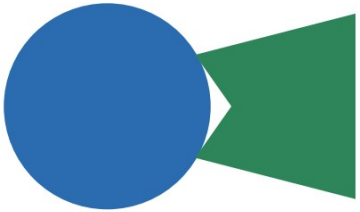
Interfaces need their own scores



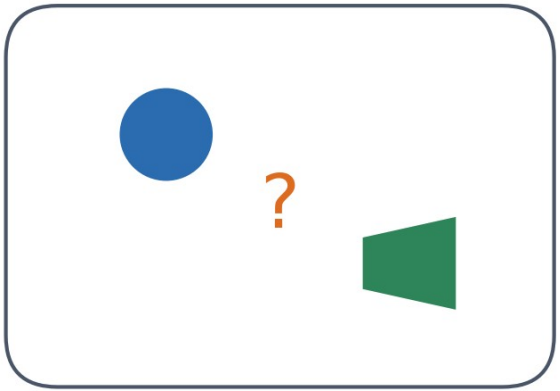
The model's own interface confidence: ipTM (AlphaFold-Multimer) or pDockQ (FoldDock: interface pLDDT and number of contacts). Against the true complex: DockQ, from 0 to 1. Each chain can be right while their placement is wrong: the PAE between chains.

Basu & Wallner (2016); Bryant et al. (2022); Evans et al. (2021).

A model shows how, not whether



HOW they could bind:
the model



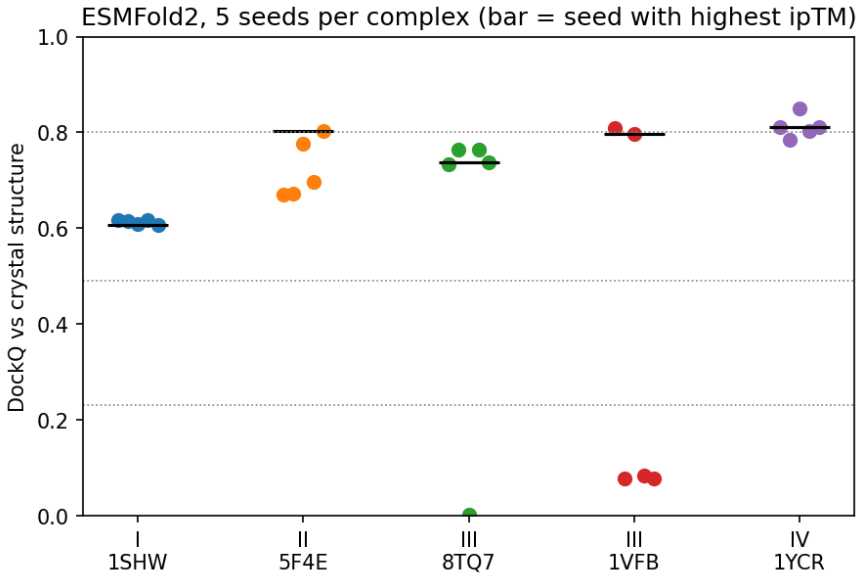
WHETHER and WHEN, in the cell

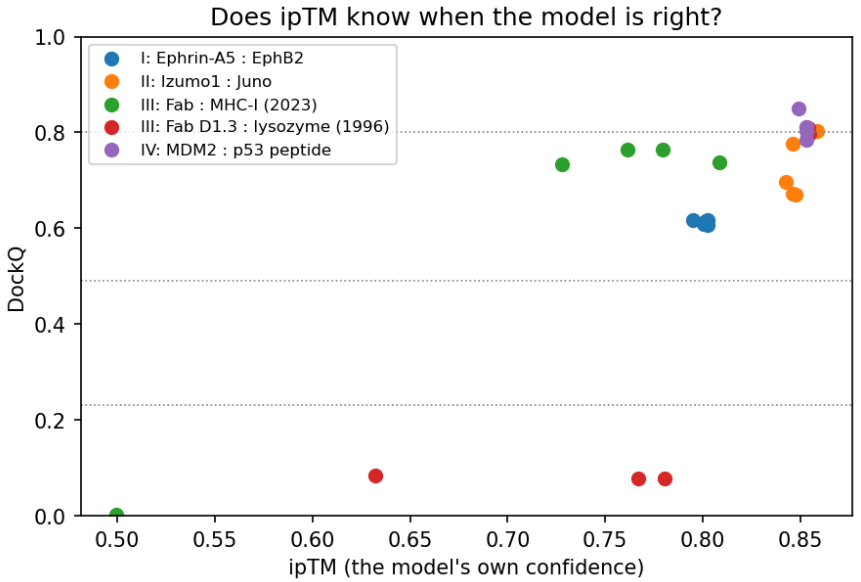
- **Plausible complex ≠ evidence of interaction**
- Combine with other evidence

Elofsson et al. (2024), eLife 13:RP93131.

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Sampling quality is not ranking quality



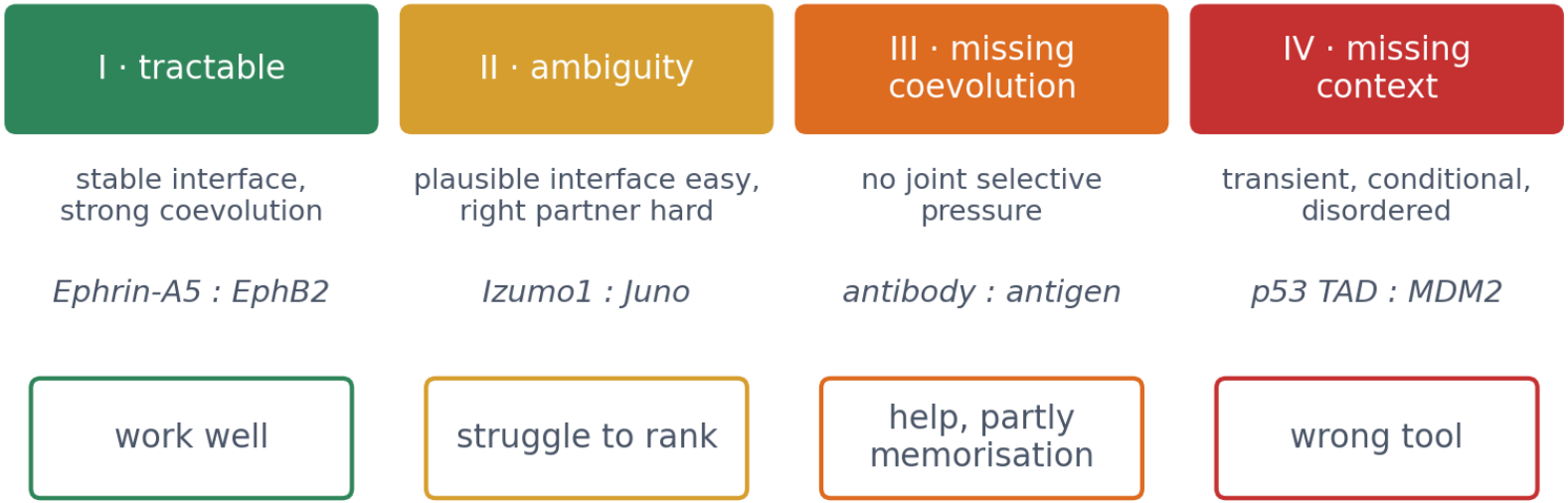


ESMFold2, five seeds per complex. The highest-ipTM seed is good for every complex, but three of five D1.3 seeds are wrong, and no single ipTM threshold separates right from wrong. Antibody-antigen: the best of 200 models reaches DockQ above 0.5, the top-ranked one is much worse.

Fromm, Ludaic & Elofsson (2026); notebook: ESMFold2, five seeds per complex.

SciLifeLab

When to trust a complex prediction: four regions



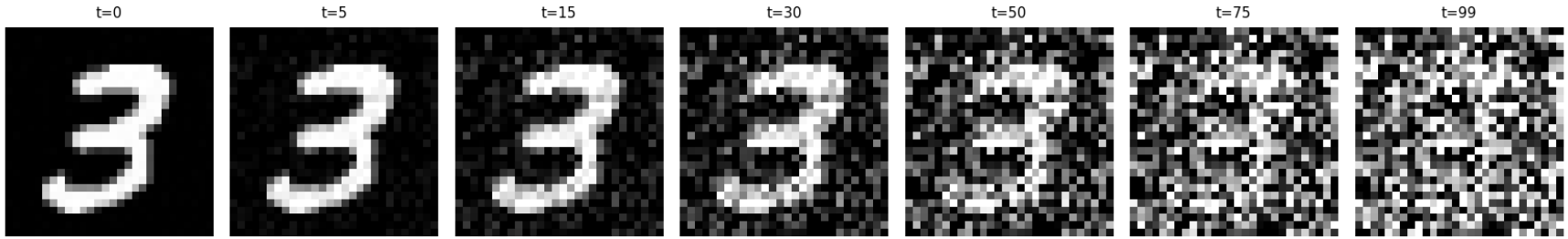
current methods → (a heuristic from the lecture, not an established classification)

A heuristic from the lecture, not an established classification. In the lab, each group predicts one complex per region on the AlphaFold Server and checks whether confidence tracks how hard the problem really is.

See course.bioinfo.se/session12.html, 'When can a complex prediction be trusted?'

SciLifeLab

Diffusion: learn to remove noise

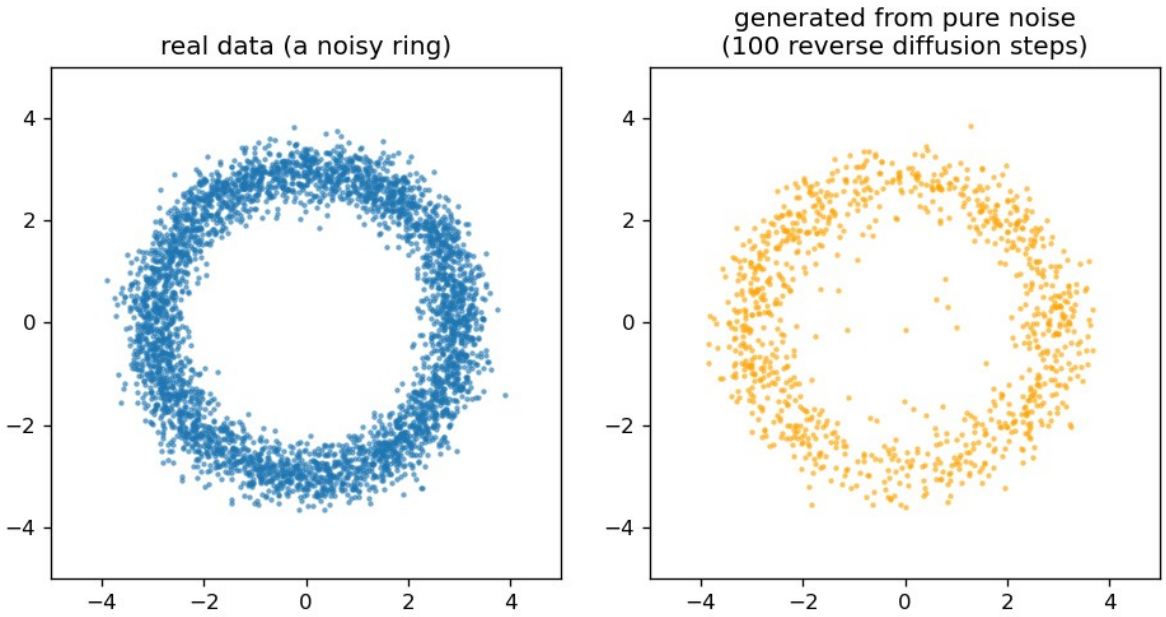


(1) Forward: add a little Gaussian noise step by step (no learning). (2) Train: predict the added noise from a noisy example and its step. (3) Generate: start from pure noise and denoise many times.

Sohl-Dickstein et al. (2015); Ho, Jain & Abbeel (2020).

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A ring from pure noise

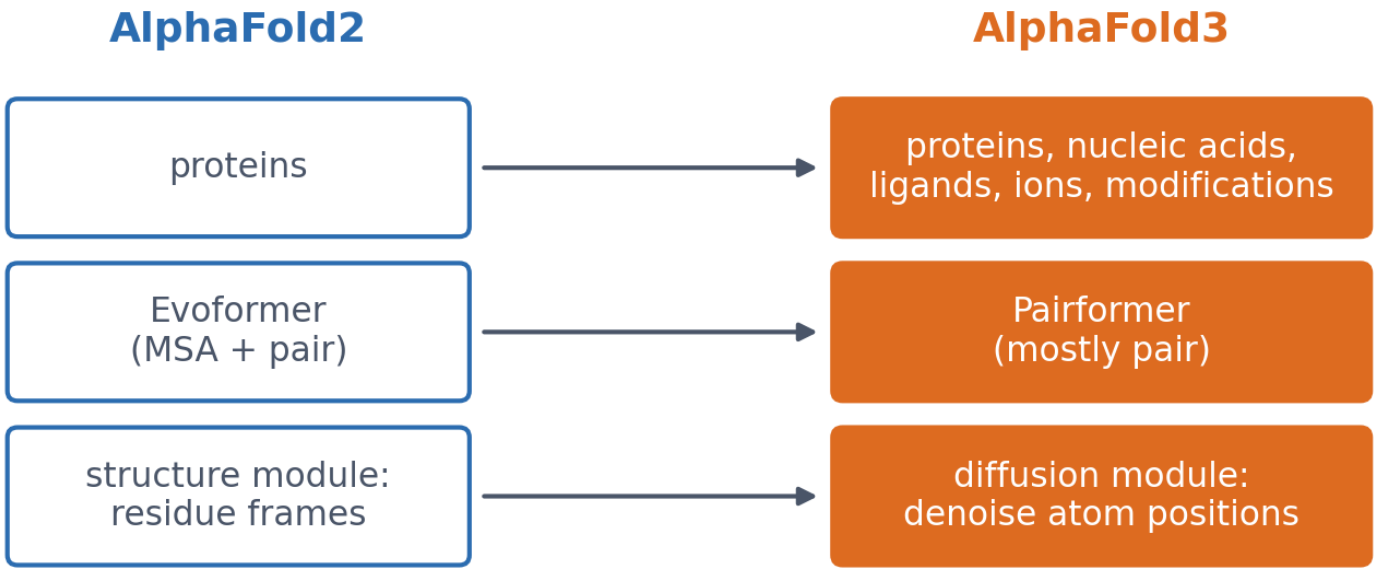


- **1,000 points generated**
- 100 denoising steps
- Learns the shape of the data

Notebook, Part 3.

SciLifeLab

AlphaFold3: what changed



All atoms at once, any molecule made of atoms. Sampling brings back the ranking problem, and the model can hallucinate structure in disordered regions. CASP16: 'slightly better for protein complexes', but the difference disappears with massive AlphaFold2 sampling.

Abramson et al. (2024), Nature 630:493; Elofsson (2026), Proteins 94:154.

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Open systems can be tested; closed ones only claimed

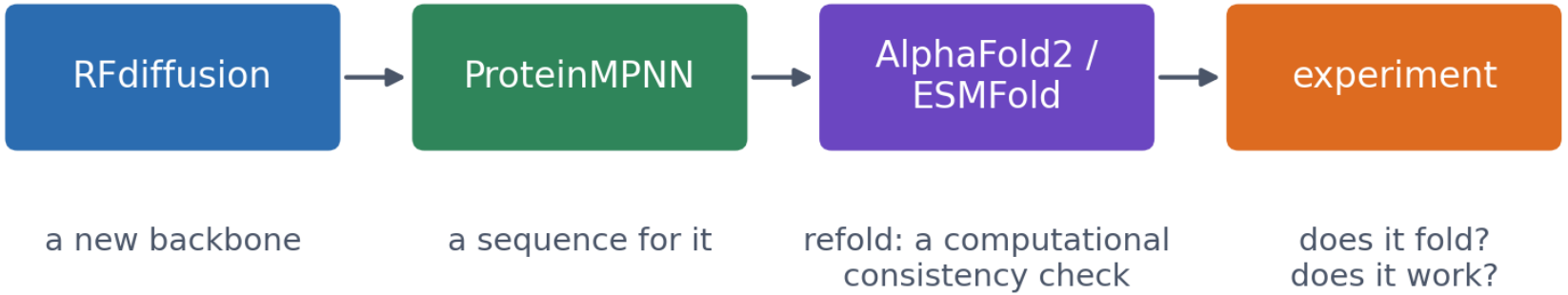
open: anyone can test them	closed: only the owners' reports
ESMFold2	IsoDDE (Isomorphic Labs)
Boltz-1, Boltz-2	no code, no model,
Chai-1, Protenix	no independent test
OpenFold3 (preview)	
OpenDDE	

AlphaFold3 was released without training code, and open reproductions followed. IsoDDE's company reports its largest gains on examples unlike the training set, where AlphaFold3 is weakest, but there is no independent test.

Optional section on the page, and it will age quickly.

SciLifeLab

Designing new proteins: one pipeline



Refolding is only a computational consistency check, not experimental validation. Hundreds of RFdiffusion designs were tested in the lab, including an influenza-haemagglutinin binder whose cryo-EM structure was 'nearly identical to the design model'.

Watson et al. (2023), Nature 620:1089; Dauparas et al. (2022), Science 378:49.

SciLifeLab

One structure, or many?

AFsample

sample the same network hard

- Dropout left ON at prediction
- ~6,000 models per target, not 25
- Top of CASP15 protein assembly

AlphaFlow

retrain AlphaFold as generative

- Flow matching
- Each run draws a different structure from a distribution

BioEmu

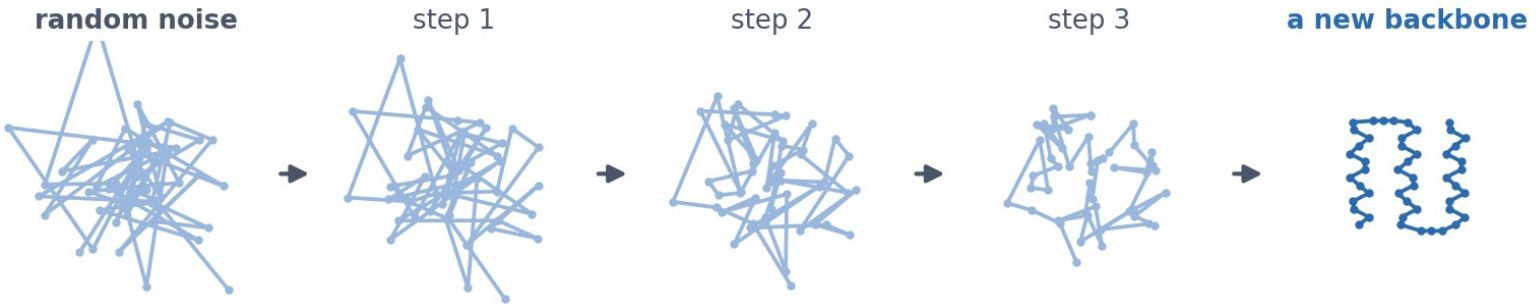
a learned shortcut for MD

- Trained on >200 ms of simulation + stabilities
- Thousands of structures per hour on one GPU
- Free energies to ~1 kcal/mol

Wallner (2023), Bioinformatics 39:btad573; Jing et al. (2024), arXiv:2402.04845; Lewis et al. (2025), Science 389:eadv9817.

SciLifeLab

RFdiffusion: a backbone out of noise



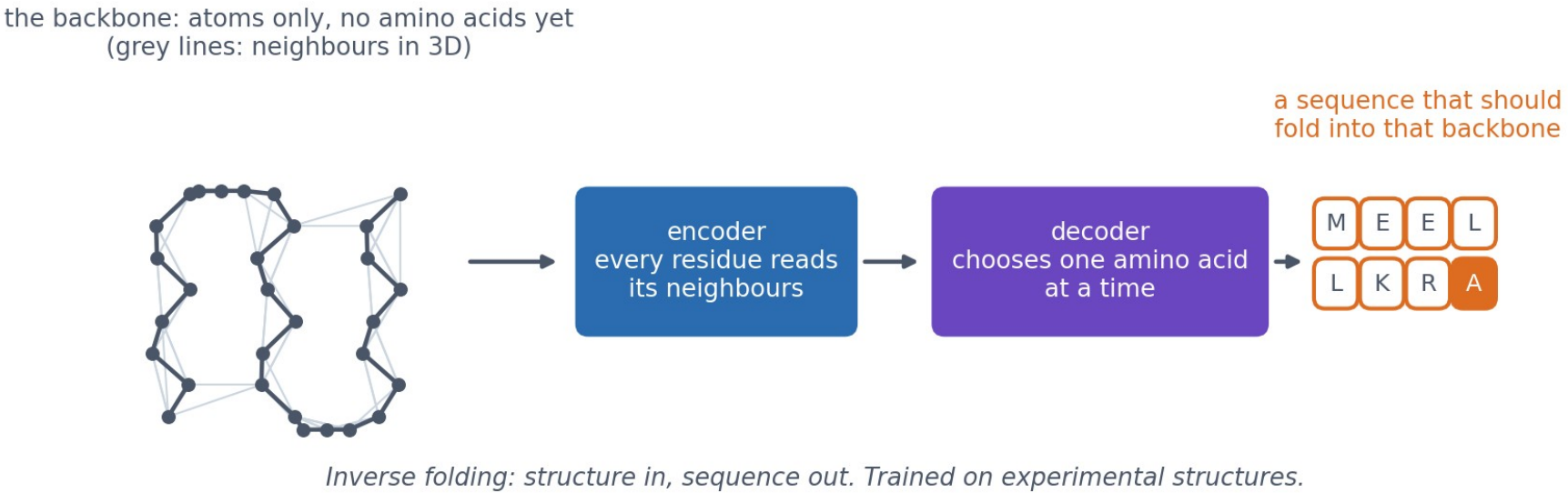
each arrow: the network removes a little noise. A target surface or a motif can be held fixed, and the rest is generated around it.

The diffusion idea of this block, applied to coordinates: start from randomly placed residues and denoise, with a fine-tuned RoseTTAFold. A target surface or a functional motif can be held fixed while the rest is generated around it.

Watson et al. (2023), Nature 620:1089. Schematic. See course.bioinfo.se/session12.html, 'Designing new proteins'.

SciLifeLab

ProteinMPNN: which sequence folds into this backbone?



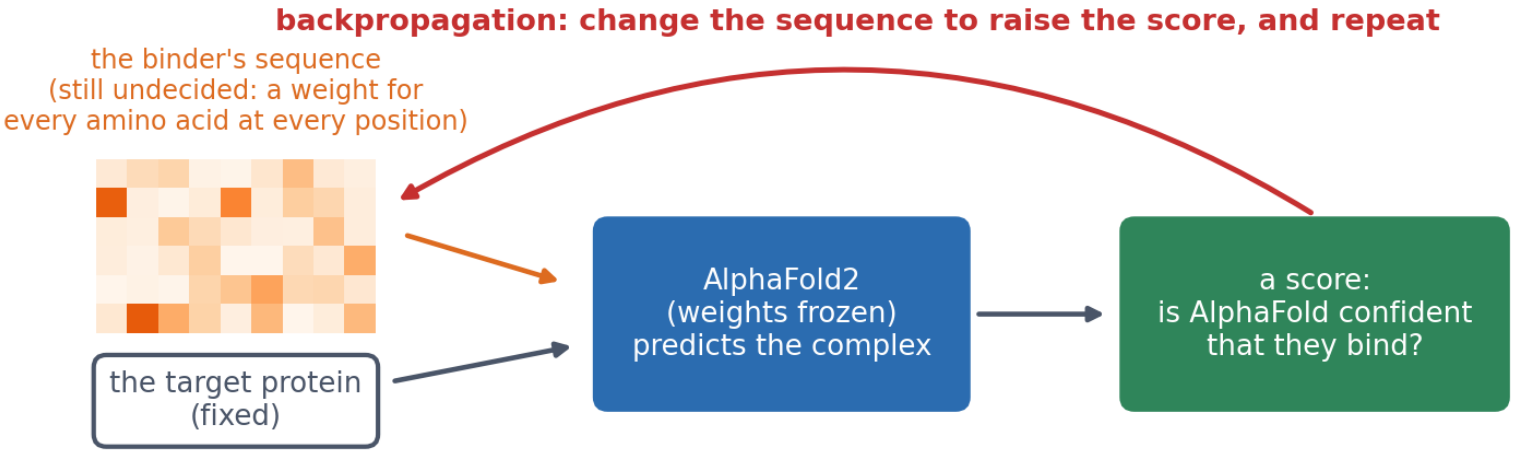
Inverse folding: structure in, sequence out. Trained on experimental structures.

Inverse folding: the structure is given, the sequence is the unknown. An encoder lets each residue read its neighbours in 3D; a decoder then chooses the amino acids one at a time, as a language model writes text. 52.4% native sequence recovery, against 32.9% for Rosetta.

Dauparas et al. (2022), Science 378:49. See course.bioinfo.se/session12.html, 'Designing new proteins'.

SciLifeLab

BindCraft: optimise the input, not the weights



The network is never trained: the gradient updates the sequence, not the weights.

AlphaFold2 is differentiable, so backpropagation (Session 9) can run in reverse: freeze the weights and change the binder's sequence until AlphaFold is confident that it binds the target. Experimental success rates of 10-100%, with no high-throughput screening.

Pacesa et al. (2025), Nature 646:483; Anishchenko et al. (2021), Nature 600:547. See course.bioinfo.se/session12.html.

SciLifeLab

From predicting structures to designing molecules

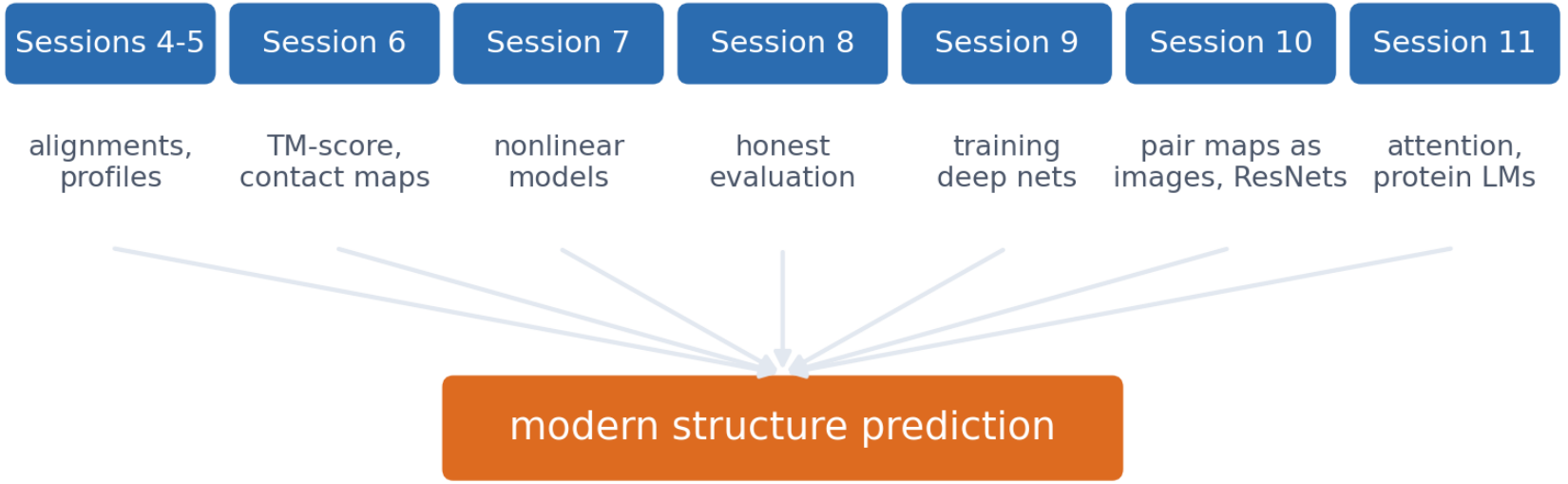
binders	RFdiffusion: designs tested in the lab; cryo-EM matches the design	Watson 2023
antibodies	de novo antibodies to chosen epitopes; cryo-EM confirms the CDRs	Bennett 2026
affinity	approaches physics-based FEP, at least 1000 times faster	Boltz-2, 2025
drugs	large gains claimed, no code, no independent test	IsoDDE, 2026

- Designed binders work in the lab
- Antibodies: designed, then matured
- Separate results from claims
- **What is left for experiments?**

Watson et al. (2023), Nature 620:1089; Bennett et al. (2026), Nature 649:183; Passaro et al. (2025), Boltz-2, bioRxiv; Isomorphic Labs (2026), IsoDDE technical report.

SciLifeLab

The course in one line



Modern structure prediction combines alignment and evolution with neural networks, attention, pair representations and geometry. A model is a hypothesis with confidence estimates, not an experiment.

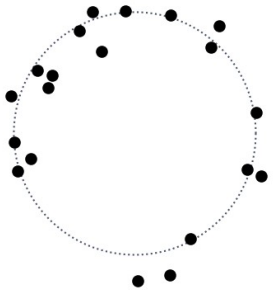
See course.bioinfo.se/session12.html, 'The course in one line'.

SciLifeLab

What does the model trained on 20 points generate?

A toy diffusion model; trained on 4000 points of a ring, it generates points spread evenly along the ring.

20 training points



noise → ring, step by step

```
1  n_train = 20
2  data = ring(n_train)
3  model = train(data)
4  gen = model.sample(1000)
```

- **A** Spread evenly, as with 4000
- **B** On the ring, bunched around the 20
- **C** Exact copies of the 20 points
- **D** Noise: 20 are too few

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

SciLifeLab

What does the model trained on 20 points generate?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What does the model trained on 20 points generate?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session12-discussion-3.ipynb

SciLifeLab

What does the model trained on 20 points generate?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

SciLifeLab

What does the model trained on 20 points generate?

Results, and who explains

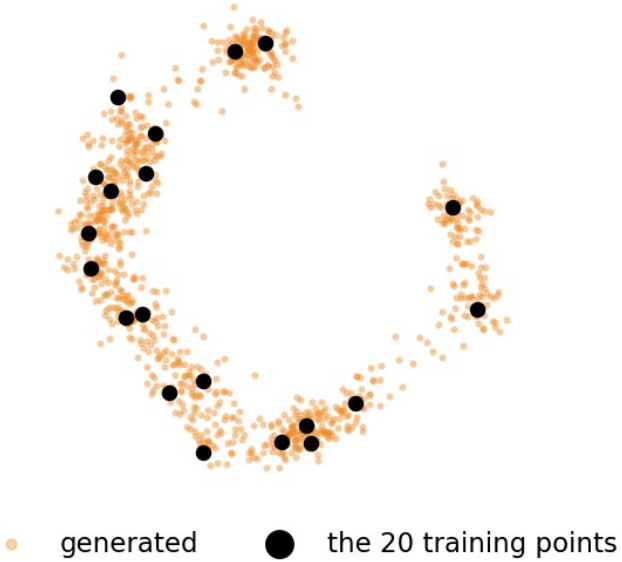
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: on the ring, bunched around the 20 examples



Median distance to the nearest training point

generated points	0.28
a perfect sampler	0.53

On the ring, but about twice as close to the examples as fresh ring points would be: bunched.

Trained on 20 points, the toy model generates points on the ring (0.04 from it), but about twice as close to the training points as a perfect ring sampler would (0.28 against 0.53): it has learned some of the shape and memorised much of the examples. AlphaFold3's high accuracy on complexes like those in its training set raises the same question; only a test set built to exclude them can answer it.

SciLifeLab

In up to three words: what was the hardest part of today?

Contacts	AlphaFold2	Complexes, design
coevolution, contact maps	Evoformer, pLDDT and PAE	interfaces, diffusion

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

SciLifeLab

The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

Next: the capstone project

Secondary structure again, with every tool from Sessions 5-12
A model is a hypothesis with confidence estimates, not an experiment